

Algorithm Benchmarking and Outcomes

**Society Connectedness & Health Lab
Yale School of Public Health**

Church Closures Dashboard Project



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Review of the 2026 Format Raw Data

Prepared By: Shelby Golden, M.S. from the Yale School of Public Health Data Science and Data Equity team

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Overview

The project objectives were as follows:

1. **REVIEW:** Evaluate the raw data and previously outlined cleaning, validation, and harmonization methods employed by Dr. Insang Song for accuracy and identify opportunities for improvement.
2. **PREPARATION:** Implement identified improvements to prepare the dataset for analysis and distribution to qualifying collaborators.
3. **METRICS:** Calculate the metrics necessary for dynamic visualization.

Additionally, the project Principal Investigator (PI), Professor Yusuf Ransome, requested the pipeline be reproducible for future collaborators and readily adaptable to potential changes in project scope.

Following receipt of the raw data in Summer 2025, a review was conducted and findings were compared against the original methodology provided by Dr. Song. When an updated dataset was subsequently delivered in May 2026, the pipeline was expanded to accommodate two designated variations: the **2023 Format** and the **2026 Format**.

This document provides an overview of the challenges associated with the 2026 Format data and identifies opportunities for validation and cleaning. Unless otherwise noted, findings outlined for the 2023 Format apply here as well; additional details are provided in the comparative review report. This report does not aim to examine every case exhaustively or fully quantify the extent of any issues — rather, it serves to characterize the data at a high level.

Important Considerations

Selected code excerpts are referenced below. The complete code used to generate this report can be found in “[Code/Process Data Update.R](#)” with all packages required documented in the **SET UP THE ENVIRONMENT** section. Custom functions used throughout the script are in “[Code/Support Functions/For Processing New Data.R](#)”.

Ignored Files

Individual-level files are withheld from GitHub and Git tracking in accordance with the Data Use Agreement (DUA) with the data provider. Many of these are stored in `**/KEEP LOCAL/` directories or are explicitly listed under the “Project-Specific Exclusions” section of the `.gitignore` file.

Most files are available on the SOCAH Lab OneDrive for the project, while others are user-specific and do not need to be shared. If there are local-only files for this project, they will be added to the “Church-Closures-Dashboard GitHub LOCAL ONLY Files” directory. This directory will contain `README.md` files specifying the intended use and filepath for each file.

If you do not have access to the SOCAH Lab OneDrive, carefully follow the codebase instructions to establish the repository and replicate results on your own device. Raw data access instructions may vary depending on your institution.

Environment Reproducibility

Results were produced using R (v4.5.2), RStudio (v2026.01.1+403), and Quarto (v1.8.27). The `renv` package (v1.1.4) is included to reproduce the coding environment, capturing all relevant packages and their versions. If issues arise when running the scripts, verify that the environment is initialized and that the correct versions of R, RStudio, and Quarto are being used.

Compatibility

Updating packages, R, or Quarto may introduce compatibility issues with legacy code that will need to be resolved on a case-by-case basis.

First-time users should initialize the environment to install all required packages and ensure a

reproducible coding environment. To do so:

1. In the command-line application (i.e., Terminal for Macs or Git Bash for Windows) navigate to the codebase file location.

Command-Line Interface

```
cd "FILE-PATH/Church-Closures-Dashboard/"
```

2. Launch the project by opening the *.Rproj file in RStudio.
3. In the R console, activate the environment by running the following lines of code:

Command-Line Interface

```
renv::restore() # Download packages and their version saved in the
↪ lockfile.
```

i Note

If you are asked to update packages, say no. `renv` is intended to recreate the same environment under which the project was created, making it reproducible. You are ready to proceed when running `renv::restore()` gives the output:

RStudio Console

```
- The library is already synchronized with the lockfile.
```

If you experience any trouble with this step, you might want to confirm that you are using R (v4.5.2) in the RStudio (v2026.01.1+403) and Quarto (v1.8.27). You can also read more about `renv` in their [vignette](#)¹.

Data Dictionary

File name: church_long_form_050926.csv

Number of rows: 10,201,197

Number of columns: 54

Some details were sourced from the Environmental Systems Research Institute (ESRI) documentation on the Data Axle dataset utilized in this study, while others were estimated based on variable naming². Some variables were neither definable from their name alone nor findable in source provider documentation.

- `company`: The business name.
- `archive_version_year`: The year the information was added, spanning 2000 to 2025.
- `abi`: American Business Identifier (ABI): The unique identifier for each business, used to associate addresses with a specific organization.
- `year_established`: Indicates the year the business was first established.
- `primary_naics8_code`: This dataset combines the six-digit 2022 vintage [North American Industry Classification System \(NAICS\)](#) code and a two-digit proprietary code from Data Axle. The two-digit Data Axle code provides additional classification details not reflected here.
- `naics8_descriptions`: All entries have NAICS = 813110, indicating they are classified as “Religious Organizations”, which includes churches, shrines, monasteries (except schools), synagogues, mosques, and temples.
- `subsidiary_number`: A unique identifier assigned to a subsidiary of the parent company, used to link the business location back to its specific subsidiary entity within the broader corporate hierarchy.
- `company_holding_status`: Indicates the holding or ownership status of the company, distinguishing between classifications such as headquarters, branch, subsidiary, or standalone entity within a corporate structure.
- `address_line_1`: The primary address line associated with the ABI entry.
- `city`: The city associated with the address.

- `state`: The state associated with the address.
- `zipcode`: The five-digit ZIP code associated with the address.
- `zip4`: The four-digit ZIP code extension associated with the address.
- `primary_sic_code`: The primary Standard Industrial Classification (SIC) code assigned to the business, representing its main industry or business activity.
- `sic6_descriptions`: Description of the SIC code.
- `sic_code_[1-4]`: Up to four additional SIC codes assigned to the business, capturing additional or secondary industries and activities beyond the primary classification.
- `sic6_descriptions_sic[1-4]`: Description of the additional SIC codes.
- `longitude` and `latitude`: Given as two columns. Provides the geolocation associated with the address listed in that row.
- `census_block`: The smallest geographic unit defined by the U.S. Census Bureau. Provides a precise sub-tract level location identifier associated with the address listed in that row.
- `census_tract`: A small, relatively permanent geographic subdivision of a county defined by the U.S. Census Bureau, used for statistical and demographic analysis.
- `county_code`: A numeric code identifying the county associated with the address listed in that row.
- `fips_code`: The Federal Information Processing Standards (FIPS) code assigned to the geographic area, providing a standardized identifier for states, counties, and other administrative regions.
- `cbsa_level`: Indicates the level of the Core-Based Statistical Area (CBSA), distinguishing between Metropolitan Statistical Areas (MSAs) and Micropolitan Statistical Areas.
- `cbsa_code`: A numeric code identifying the Core-Based Statistical Area (CBSA) associated with the address, as defined by the U.S. Office of Management and Budget (OMB).
- `csa_code`: A numeric code identifying the Combined Statistical Area (CSA) associated with the address, representing a grouping of adjacent CBSAs with strong economic and

social ties.

- `area_code`: The telephone area code associated with the business location.
- `site_number`: A unique identifier assigned to a specific physical location or site of a business, used to distinguish individual locations within the same organization.
- `yellow_page_code`: A code indicating the Yellow Pages heading(s) under which the business is listed, reflecting how the business advertises or categorizes itself in directory listings.
- `office_size_code`: A coded value representing the size of the office or physical location.
- `employee_size_location` and `location_employee_size_code`: Given as two columns. Provides the actual employee count at the specific location and a corresponding code representing that count within a size classification range.
- `sales_volume_location` and `location_sales_volume_code`: Given as two columns. Provides the actual sales attributed to the specific location and a corresponding code representing that number within a size classification range.
- `parent_number`: A unique identifier assigned to the parent company of the business, used to link individual locations or subsidiaries back to their controlling organization.
- `parent_actual_employee_size` and `parent_employee_size_code`: Given as two columns. Provides the total employee count for the parent company and a corresponding code representing that count within a size classification range.
- `parent_actual_sales_volume` and `parent_sales_volume_code`: Given as two columns. Provides the total sales for the parent company and a corresponding code representing that number within a size classification range.
- **Unknown variables or code translations:** `address_type_indicator`, `industry_specific_first_byte`, `business_status_code`, `population_code`, `match_code`, `ticker`, and `idcode`.

Correspondence to the 2023 Format

RStudio Console

```
# Results from "PART A: ASSESS METADATA CONTENT DIFFERENCES" found in
# `Process Data Update.R`

handling_var <- read.csv("../Data/Results/From Process Data
  ↪ Update/Handling Raw Variables_05.12.2026.csv") %>%
  as.data.frame()
```

Core fields (abi, address_line_1, city, and state) are consistent between the old and new data formats. Not shown here, the 2026 Format introduces the variable `archive_version_year`, which appends new years of information as rows, whereas the 2023 Format extended this information column-wise as binary encodings. Notably, the new data stores the zip code as a five-digit character field, preserving leading zeros, and includes a four-digit zip code extension.

Table 1: Variable classes and metadata flag for 2026 vs. 2023.

#	Variable	Metadata Var	Class_2026	Class_2023
1	company	TRUE	character	<NA>
2	abi	FALSE	numeric	numeric
3	year_established	TRUE	numeric	numeric
4	primary_naics6_code	TRUE	numeric	numeric
5	naics6_descriptions	TRUE	character	character
6	subsidiary_number	TRUE	character	<NA>
7	company_holding_status	TRUE	numeric	<NA>
8	address_line_1	FALSE	character	character
9	city	FALSE	character	character
10	state	FALSE	character	character
11	zipcode	FALSE	character	numeric
12	zip4	FALSE	character	<NA>
13	primary_sic_code	TRUE	numeric	<NA>
14	sic6_descriptions	TRUE	character	<NA>
15	sic_code	TRUE	numeric	<NA>

#	Variable	Metadata Var	Class_2026	Class_2023
16	sic6_descriptions_sic	TRUE	character	<NA>
17	sic_code_1	TRUE	numeric	<NA>
18	sic6_descriptions_sic1	TRUE	character	<NA>
19	sic_code_2	TRUE	numeric	<NA>
20	sic6_descriptions_sic2	TRUE	character	<NA>
21	sic_code_3	TRUE	numeric	<NA>
22	sic6_descriptions_sic3	TRUE	character	<NA>
23	sic_code_4	TRUE	logical	<NA>
24	sic6_descriptions_sic4	TRUE	logical	<NA>
25	latitude	FALSE	numeric	numeric
26	longitude	FALSE	numeric	numeric
27	census_block	FALSE	numeric	<NA>
28	census_tract	FALSE	character	<NA>
29	county_code	FALSE	character	<NA>
30	fips_code	FALSE	character	<NA>
31	cbsa_level	FALSE	numeric	<NA>
32	cbsa_code	FALSE	character	<NA>
33	csa_code	FALSE	character	<NA>
34	area_code	FALSE	numeric	<NA>
35	site_number	TRUE	character	<NA>
36	yellow_page_code	TRUE	numeric	<NA>
37	office_size_code	TRUE	logical	<NA>
38	employee_size_location	TRUE	numeric	<NA>
39	location_employee_size_code	TRUE	character	<NA>
40	sales_volume_location	TRUE	numeric	<NA>
41	location_sales_volume_code	TRUE	character	<NA>
42	parent_number	TRUE	character	<NA>
43	parent_actual_employee_size	TRUE	numeric	<NA>
44	parent_employee_size_code	TRUE	character	<NA>
45	parent_actual_sales_volume	TRUE	numeric	<NA>
46	parent_sales_volume_code	TRUE	character	<NA>
47	primary_naics2_code	TRUE	numeric	numeric
48	address_type_indicator	TRUE	character	<NA>
49	industry_specific_first_byte	TRUE	character	<NA>
50	business_status_code	TRUE	numeric	<NA>
51	population_code	TRUE	numeric	<NA>
52	match_code	TRUE	character	<NA>
53	ticker	TRUE	logical	<NA>

#	Variable	Metadata Var	Class_2026	Class_2023
54	idcode	TRUE	numeric	<NA>

The new dataset also introduces additional columns covering census and location attributes, as well as business-level details including size, status, and SIC encoding. Because the data structure appends new information row-wise, care must be taken to ensure these metadata fields are correctly collapsed when the data is transformed into single rows per ABI and address. Ideally, the structural transformation should occur after addresses have been grouped and matched to similar ones, to reduce redundancies introduced by typographical errors.

Variables can be categorized as follows:

Table 2: Variable grouping classifications.

#	Variable	Group
1	company	Core business details
2	abi	Core business details
3	year_established	Core business details
4	primary_naics6_code	Core business details
5	naics6_descriptions	Core business details
6	subsidiary_number	Core business details
7	company_holding_status	Core business details
8	address_line_1	Core business details by location
9	city	Core business details by location
10	state	Core business details by location
11	zipcode	Core business details by location
12	zip4	Core business details by location
13	primary_sic_code	Core business details by location
14	sic6_descriptions	Core business details by location
15	sic_code	Core business details by location
16	sic6_descriptions_sic	Core business details by location
17	sic_code_1	Core business details by location
18	sic6_descriptions_sic1	Core business details by location
19	sic_code_2	Core business details by location
20	sic6_descriptions_sic2	Core business details by location
21	sic_code_3	Core business details by location
22	sic6_descriptions_sic3	Core business details by location
23	sic_code_4	Core business details by location

#	Variable	Group
24	sic6_descriptions_sic4	Core business details by location
25	latitude	Core location details
26	longitude	Core location details
27	census_block	Core location details
28	census_tract	Core location details
29	county_code	Core location details
30	fips_code	Core location details
31	cbsa_level	Core location details
32	cbsa_code	Core location details
33	csa_code	Core location details
34	area_code	Core location details
35	site_number	Core business details by location
36	yellow_page_code	Core business details by location
37	office_size_code	Core business details by location
38	employee_size_location	Core business details by location
39	location_employee_size_code	Core business details by location
40	sales_volume_location	Core business details by location
41	location_sales_volume_code	Core business details by location
42	parent_number	Parent company details
43	parent_actual_employee_size	Parent company details
44	parent_employee_size_code	Parent company details
45	parent_actual_sales_volume	Parent company details
46	parent_sales_volume_code	Parent company details
47	primary_naics2_code	Unknown variables or code translations
48	address_type_indicator	Unknown variables or code translations
49	industry_specific_first_byte	Unknown variables or code translations
50	business_status_code	Unknown variables or code translations
51	population_code	Unknown variables or code translations
52	match_code	Unknown variables or code translations
53	ticker	Unknown variables or code translations
54	idcode	Unknown variables or code translations

By this classification, there are two types of metadata:

- a) Metadata relevant to a given business or location that does not change across years (e.g. `year_established`, `fips_code`)

b) Metadata that may change across years (e.g. `census_block`, `employee_size_location`)

This analysis requires only most of the core business and location details. The remaining metadata will be processed simultaneously to produce a final, cleaned dataset, though not with any specific follow-up analysis in mind. As a result, the metadata may not be in a format suitable for all analyses, as the data are primarily prepared for the dashboard.

Metadata are processed by considering change across location (distinguishing between values consistent for a given business and those that vary by location) and year. For both types of location variation, value changes by year are captured to maintain granularity. While these are not expected to indicate anything actionable, they may reveal opportunities for standardization or data cleaning and validation. Some of these opportunities will be taken here, while the remainder will be left for follow-up analyses.

To capture year-level granularity in value changes, a date range will be added to each unique value when collapsed by `abi` and `combined_address` (a formatted combination of all address components). For fields that are relevant to this project, unique values that do not span the expected years — the years in which that `abi` and address were recorded — will be cleaned to report only the distinct value.

In the following sections, specific metadata fields are assessed for quality and to develop this condensing approach.

NAICS and SIC Encodings

NAICS and SIC encodings provide additional context about the type of business, religious affiliation, and function. The 2023 Form of the data only included the NAICS encoding, while the 2026 Form includes both.

Primary NAICS and SIC Encodings

When processing the 2023 Formatted data, it was identified that the NAICS code is eight digits long, with the trailing two digits representing an unknown proprietary code from Data Axle. These will be separated into a six-digit (`naics6`) code with description and a two-digit (`naics2`) code with no associated description.

Both versions of the data share the same unique NAICS eight-digit encodings. The 2023

Formatted data processing protocol retained only the unique six-digit NAICS codes and their descriptions, which are consistent across all entries; the trailing two digits were stored as a string. The method devised to retain applicable years will also be applied to this field.

SIC refines the classification of religious organizations by capturing attributes such as affiliation, denomination, and operational function (e.g., school, retreat center). This information was not provided in the previous version. It is provided as both a primary classification column accompanied by up to four additional overflow columns. All are examined together in the following section.

Primary and Additional SIC Encodings

The following assumptions about the SIC encodings require review and confirmation:

- a. Overflow columns are filled progressively with no other discernible pattern.
- b. The same categories appear across all columns, with the possible exception of the primary SIC column, which may differ or be restricted.
- c. Each code maps to a unique description and vice versa.
- d. Nomenclature is consistent throughout.

RStudio Console

```
# Custom SIC column auditing function devised to quickly assess the above
# assumptions for recently exported data in the 2026 Format.
# ~ line 415

sic_results <- sic_overflow_audit(church_long_form_050926)
```

! Important

The following results apply to the most recent data export in the 2026 Format at the time of this writing. To ensure subsequently exported data in this format are appropriately processed, they will need to be reaudited to confirm the validity of these assumptions or

flag any corrections that are required.

ASSUMPTION A: Each additional SIC encoding is assumed to represent supplementary classification information attributed to that address. The code columns follow the expected pattern; however, the description columns do not. This is likely due to missing descriptions for certain codes. The `sic_results$missing_desc_codes_are_subset_of_no_desc = TRUE` result confirms that description fields deviating from this pattern do so because of missing descriptions for a given code.

RStudio Console

```
# Select custom SIC column auditing function results.
# ~ line 418-436

sic_results$overflow_summary_code
```

Total Rows	Rows Passed	Rows Failed
10,201,197	10,201,197	0

RStudio Console

```
# Select custom SIC column auditing function results.
# ~ line 418

sic_results$missing_desc_codes_are_subset_of_no_desc # TRUE
sic_results$overflow_summary_desc
```

Total Rows	Rows Passed	Rows Failed
10,201,197	10,201,098	99

ASSUMPTION B: If each column represents an overflow of the same underlying information, a significant degree of overlap in the values present across columns would be expected.

RStudio Console

```
# Select custom SIC column auditing function results.
# ~ line 437-445

sic_results$presence_wide      # All SIC codes/descriptions
sic_results$presence_tabs      # Tabulated outcomes
```

Row	FALSE	TRUE
primary_sic_code + sic6_descriptions	2,563	43
sic_code + sic6_descriptions_sic	353	2,253
sic_code_1 + sic6_descriptions_sic1	1,149	1,457
sic_code_2 + sic6_descriptions_sic2	1,652	954
sic_code_3 + sic6_descriptions_sic3	1,984	622

There is a high degree of overlap across the overflow SIC encodings, which decreases as the overflow index increases. This is expected, as fewer entries carry more than one or two additional SIC classifications.

Only a small subset of possible SIC codes is utilized in the primary column, suggesting that the available selection may be more restricted at the point of data collection. All 43 outcomes in the primary SIC are also used in overflow SIC columns.

RStudio Console

```
# Select custom SIC column auditing function results.
# ~ line 456-466

# Pull columns that are logical TRUE/FALSE "sic presence" columns
sic_cols <- sic_results$presence_wide %>%
  select(where(is.logical)) %>%
  names()
```

```

other_cols <- setdiff(sic_cols, "primary_sic_code + sic6_descriptions")

# All Primary SIC outcomes are present in the overflow categories
sic_results$presence_wide %>%
  filter(.data[["primary_sic_code + sic6_descriptions"]] &
    ↪ if_any(all_of(other_cols), ~ .x)) %>%
  nrow()      # 43

```

ASSUMPTION C: It is assumed that each code maps uniquely to a single description and vice versa. To verify this assumption, a mapping table is constructed from all non-null (code, description) pairs across all column pairs. Consistency checks:

- a) Each SIC code maps to exactly one description.
- b) Each SIC description maps to exactly one code.
- c) SIC codes associated with an NA description are identified and accounted for elsewhere.

RStudio Console

```

# Select custom SIC column auditing function results.
# ~ line 473-487

sic_results$map_tbl

other_cols <- setdiff(sic_cols, "primary_sic_code + sic6_descriptions")

list(
  n_unique_pairs = sic_results$n_unique_pairs,
  code_to_desc_consistent = sic_results$code_to_desc_consistent,
  desc_to_code_consistent = sic_results$desc_to_code_consistent,
  codes_with_multiple_desc = sic_results$codes_with_multiple_desc,
  desc_with_multiple_codes = sic_results$desc_with_multiple_codes,
  desc_that_is_sometimes_na = sic_results$desc_that_is_sometimes_na
)

```

Summary

Unique pairs	2,583
Code-to-desc consistent	FALSE
Desc-to-code consistent	FALSE

Codes with Multiple Descriptions

SIC Code	N Descriptions
279610	2
513705	2
549903	2
561101	2
562101	2
596316	2
799922	2
799927	2
804908	2
822298	2
824401	2
824903	2
835102	2
864102	2
869904	2
874881	2
912101	2
912104	2

Descriptions with Multiple Codes

SIC Description	N Codes
Computer Software	2
Schools—Business & Secretarial	2

Descriptions That Are Sometimes NA

SIC Code	SIC Description
866124	Monasteries
821120	School Districts
737206	Computer Software

Some codes are associated with multiple descriptions and vice versa. Additionally, some codes are sometimes associated with an NA description despite having a valid description in other instances. Each case will be addressed individually.

ASSUMPTION D: Many of these inconsistencies are attributable to variations in nomenclature and will be corrected on an individual basis.

RStudio Console

```
# Entries to correct description
# ~ line 498

fix_desc <- sic_results$presence_wide %>%
  filter(sic_code %in% sic_results$codes_with_multiple_desc$sic_code) %>%
  select(sic_code, sic_desc) %>%
  arrange(sic_code) %>%
  group_by(sic_code) %>%
  mutate(dup_id = row_number()) %>%
  ungroup() %>%
  pivot_wider(
    id_cols = sic_code,
    names_from = dup_id,
    values_from = sic_desc,
    names_prefix = "sic_desc_"
  ) %>%
  as.data.frame()
```

Table 3: Reduplicated SIC code descriptions.

sic_code	sic_desc
279610	PRINTERS SERVICES (MFRS)
279610	PRINTERS' SERVICES (MFRS)
513705	WOMEN'S APPAREL-WHOLESALE
513705	WOMENS APPAREL-WHOLESALE
549903	WATER COMPANIES-BOTTLED, BULK, ETC
549903	WATER COMPANIES-BOTTLED/BULK & ETC
561101	MEN'S CLOTHING & FURNISHINGS-RETAIL
561101	MENS CLOTHING & FURNISHINGS-RETAIL
562101	WOMEN'S APPAREL-RETAIL
562101	WOMENS APPAREL-RETAIL
596316	COFFEE & FOOD SVC-MOBILE
596316	MOBILE CONCESSIONS
799922	NATIONAL LANDMARK
799922	NATIONAL LANDMARK/HISTORIC SITE/MEMORIAL
799927	CHILDREN'S SVCS & ACTIVITIES INFORMATION
799927	CHILDRENS SVCS & ACTIVITIES INFORMATION
804908	NURSES & NURSES REGISTRIES
804908	NURSES & NURSES' REGISTRIES
822298	JUNIOR COLLEGES & TECHNICAL INSTITUTES
822298	JUNIOR-COMMUNITY COLLEGE-TECH INSTITUTES
824401	SCHOOLS-BUSINESS & SECRETARIAL
824401	SCHOOLS-BUSINESS & VOCATIONAL
824903	SCHOOLS-INDUSTRIAL TECHNICAL & TRADE
824903	SCHOOLS-TRADE & VOCATIONAL
835102	SCHOOLS-NURSERY & KINDERGARTEN ACADEMIC
835102	SCHOOLS-PRE-SCHOOL/KINDERGARTEN-ACADEMIC
864102	VETERANS & MILITARY ORGANIZATIONS
864102	VETERANS' & MILITARY ORGANIZATIONS
869904	WOMEN'S ORGANIZATIONS & SERVICES
869904	WOMENS ORGANIZATIONS & SERVICES
874881	EDUCATIONAL ASSESSEMENT
874881	EDUCATIONAL ASSESSMENT
912101	GOVERNMENT OFFICES-FEDERAL
912101	GOVERNMENT OFFICES-US
912104	GOVERNMENT OFFICES-CITY, VILLAGE & TWP
912104	GOVERNMENT OFFICES-CITY/VILLAGE & TWP

RStudio Console

```
# Entries to correct code
# ~ line 514

fix_code <- sic_results$presence_wide %>%
  semi_join(
    sic_results$presence_wide %>%
      filter(sic_desc %in% sic_results$desc_with_multiple_codes$sic_desc) %>%
      distinct(sic_code),
    by = "sic_code"
  )
```

Table 4: SIC descriptions with different codes (subset; additional columns omitted).

sic_code	sic_desc	primary_sic_code + sic6_desc...	sic_code + sic6_desc...
573401	COMPUTER SOFTWARE	FALSE	TRUE
737206	NA	FALSE	TRUE
737206	COMPUTER SOFTWARE	FALSE	TRUE
824401	SCHOOLS-BUSINESS & SEC...	FALSE	TRUE
824401	SCHOOLS-BUSINESS & VOC...	FALSE	TRUE
824402	SCHOOLS-BUSINESS & SEC...	FALSE	TRUE

RStudio Console

```
# Entries to correct NA
# ~ line 522

fix_na <- sic_results$presence_wide %>%
  semi_join(
    sic_results$presence_wide %>%
      filter(sic_desc %in% sic_results$desc_that_is_sometimes_na$sic_desc)
      ↪ %>%
      distinct(sic_code),
    by = "sic_code"
  )
```

Table 5: SIC codes associated with a non-NA description that is sometimes NA (subset; additional columns omitted).

sic_code	sic_desc	primary_sic_code + sic6_de...	sic_code + sic6_desc...
573401	COMPUTER SOFTWARE	FALSE	TRUE
737206	NA	FALSE	TRUE
737206	COMPUTER SOFTWARE	FALSE	TRUE
821120	NA	FALSE	TRUE
821120	SCHOOL DISTRICTS	FALSE	TRUE
866124	NA	TRUE	TRUE
866124	MONASTERIES	TRUE	TRUE

These SIC code characteristics are expected to vary with additional years of data or subsequent exports of previous reports, as Data Axle may update their database over time. It will therefore be important to process these columns from a raw data export prior to beginning any analysis or data validation.

After reconciling nomenclature variations, unique SIC values for a given ABI and address will be retained and expanded into overflow columns. The primary SIC encoding will be constrained to align with qualifying classifiers, while any additional SIC columns will capture all other unique SIC values present.

Handling Additional Metadata

In the previous section, two encodings were examined: NAICS and SIC. NAICS remained uninformative, while SIC provided additional important functional and affiliation details (described above). The remaining metadata fall into two handling categories:

- Metadata not directly relevant to the current project. These fields will be retained with no effort made to clean or validate the values.
- Metadata supplying additional location details, such as county code, census block, and FIPS codes. These fields will be assessed for validation in subsequent steps while being handled similarly to category (a). Note that attributed census block and related fields may not translate directly to specific decennial maps, potentially necessitating reference to an external database.

As described above, metadata can vary by location and across the years a business was recorded. As no cleaning or validation effort is being applied to category (a), both location and year-level variation will be retained. The same approach will apply to category (b), with the addition of Boolean validation checks to capture:

- Whether the same location outcome spans the complete expected date range.
- Whether multiple location outcomes are associated with the same address.
- Whether multiple identified location outcomes vary by decennial period or follow alternative patterns³.

Refer to [Appendix - summarize_many_code_ranges_dt\(\) Output](#) for information on where to find the code that generates these results, as well as the results codebook.

RStudio Console

```
# Custom metadata column collapsing function that retains observed date
# ranges for a given ABI and address combination.
# ~ line 583-659
summarize_many_code_ranges_dt(df, vars)

# Generate a subset to test the algorithms on
subset <- church_long_form_050926 %>%
  filter(abi %in% unique(church_long_form_050926$abi)[6:10]) %>%
  mutate(
    across(c(census_block, cbsa_level, area_code), as.character),
    combined_address = paste(
      address_line_1, city, state,
      if_else(!is.na(zip4) & zip4 != "",
        paste0(zipcode, "-", zip4),
        as.character(zipcode)
      ),
      sep = ", "
    )
  ) %>%
  relocate(combined_address, .after = zip4)
```

! Important

Results shown are simulated to reflect characteristics observed in the data.

Supplementary Location Metadata

RStudio Console

```
# SUBSECTION C1: Supplementary Location Metadata
# ~ line 611-618

vars_for_loc <- c(
  # Core business location details (8)
  "census_block", "census_tract", "county_code", "fips_code", "cbsa_level",
  "cbsa_code", "csa_code", "area_code"
)

loc_condensed <- summarize_many_code_ranges_dt(subset, vars_for_loc) %>%
  arrange(abi)
```

Table 6: Census tract range checks (split to fit within margins; subset shown).

#	available_year_ranges	census_tract_ranges
1	1998-2001	101020 (1998-2001)
2	2009-2010	204999 (2009-2010)
3	2014-2017	330120 (2014-2017)
4	1999-2006, 2011-2013, 2018-2025	900001 (1999-2002), 900010 (2003-2007), 900010 (2010-2013), 900015 (2018-2025)
5	2001-2025	777701 (2001-2012), 777709 (2011-2025)
6	1997-2019	000000 (1999), 045600 (2000-2009), 045601 (2010-2019)
7	2006-2012, 2016-2025	512340 (2006-2009), 512341 (2010-2012), 512341 (2016-2025)

#	census_tract _ranges_ok	census_tract _ranges_decade_ok	census_tract _values_seen	census_tract _missing_years	census_tract _conflict_years
1	TRUE	NA	101020	NA	NA
2	TRUE	NA	204999	NA	NA
3	TRUE	NA	330120	NA	NA
4	FALSE	FALSE	900001, 900010, 900015	2008-2009	NA
5	FALSE	FALSE	777701, 777709	NA	2011-2012
6	FALSE	TRUE	000000, 045600, 045601	NA	NA
7	FALSE	TRUE	512340, 512341	NA	NA

Looking at the results for the `census_tract` variable, nearly half of the entries failed the consistency check and did not pass the check confirming that updates were made over decennial periods, as would be expected. This may be correct or could indicate broader data accuracy problems; either way, it points to notable inconsistencies that would be challenging to reconcile given the available results.

This assessment will be incorporated when condensing the full dataset. However, these findings suggest that the additional location values will need to be freshly associated using a trusted database rather than relying on the provided information.

Metadata Not Contributive to Project Goals

The following illustrates the process of condensing additional other metadata into one line per unique ABI and address combination. Note that `primary_naics2_code` is not represented here, as it will be derived from `primary_naics8_code` in the main dataset. This field falls under the “Unknown Variables or Code Translations” category.

RStudio Console

```
# SUBSECTION C2: Metadata Not Contributive to Project Goals
# ~ line 669-685

vars_to_sum <- c(
  # Core business details by business ID (4):
  "company", "year_established", "subsidiary_number",
  ↪ "company_holding_status",
```

```

# Core business details by location and might change by year (7):
"site_number", "yellow_page_code", "office_size_code",
↪ "employee_size_location",
"location_employee_size_code", "sales_volume_location",
↪ "location_sales_volume_code",
# Parent company details, where some might change by year (5):
"parent_number", "parent_actual_employee_size",
↪ "parent_employee_size_code",
"parent_actual_sales_volume", "parent_sales_volume_code",
# Unknown variables or code translations (8):
#primary_naics2_code
"address_type_indicator", "industry_specific_first_byte",
"business_status_code", "population_code", "match_code", "ticker", "idcode"
)

subset_condensed <- summarize_many_code_ranges_dt(subset, vars_to_sum) %>%
  arrange(abi)

```

Table 7: Range-collapsed metadata fields (split to fit within margins; subset shown).

#	available_year _ranges	employee_size_location _ranges
1	1999–2001	1 (1999–2001)
2	2002–2004	2 (2002–2004)
3	2008–2010	7 (2008–2010)
4	2011–2014	6 (2011–2012), 4 (2013–2014)
5	2006–2012, 2016–2018, 2022–2025	3 (2006–2012), 5 (2016–2018), 9 (2022–2025)
6	2000–2025	8 (2000–2009), 9 (2010–2025)

#	location_employee_size_code_ranges	business_status_code_ranges	population_code_ranges
1	A (1999–2001)	1 (1999–2001)	2 (1999–2001)
2	A (2002–2004)	2 (2002–2004)	8 (2002–2004)
3	C (2008–2010)	4 (2008–2010)	3 (2008–2010)
4	C (2011–2012), B (2013–2014)	4 (2011–2014)	6 (2011–2014)
5	B (2006–2012), C (2016–2018), D (2022–2025)	7 (2006–2012), 7 (2016–2018), 3 (2022–2025)	1 (2006–2008), 4 (2009–2012), 6 (2016–2018), 5 (2022–2025)
6	D (2000–2025)	9 (2000–2025)	7 (2000–2007), 6 (2008–2015), 5 (2016–2025)

Concluding Thoughts

This document outlined additional insights and data processing requirements identified in the 2026 Format. A condensing framework is provided to collapse metadata fields that are not relevant to the project or are of questionable quality. This method retains unique values along with the years in which they were recorded, allowing analysts to preserve this information for follow-up data quality checks, validation, and cleaning.

Metadata Handling Summary

Rows will be condensed by unique `abi` and `combined_address`. NAICS codes will be split into the six leading digits and a trailing two-digit proprietary code, with the latter lacking a description. The six-digit section is consistent across all entries, while the trailing two digits will be condensed by unique value and observed date range.

SIC codes will be cleaned to ensure consistency and reconcile unexpected missing values or duplications. After condensing, the primary SIC codes will be organized first, followed by any additional overflow values detected across the years the ABI and address are recorded.

Business location will continue to be primarily attributed to the given address and longitude/latitude values. The assessment of the additional provided location fields presented here reveals unexpected inconsistencies that would be difficult or impractical to reconcile. For this reason, it is recommended that decennial location information be associated

to each entry by geolocation, as outlined in the 2023 Form processing methods.

In addition to the above, data cleaning insights gleaned from the 2023 Format are assumed to apply to this version as well.

Step 1 Algorithm Updates

The 2026 Format begins in a different structure (long vs. wide), necessitating some changes to the initial data cleaning and validation steps while also providing new opportunities that impact the step-wise framework devised for the 2023 Format.

Prior to transforming the data from long form (years as rows) to wide form (years as columns), addresses will be validated against the USPS database and assessed for similarity — first matching to a validated address, then only to reduce unnecessary variations due to typographical errors. Following the prototype implementation of **Step 1** outlined for the 2023 Format, two issues were identified: years-open and years-closed values were mistakenly duplicated, and some condensed addresses failed the geolocation similarity check.

To avoid inducing these errors, string-matched addresses will be triaged based on qualifying criteria, with one top candidate selected. These will then be joined to the whole table by ABI and compiled address, rather than `address_line_1` alone, to ensure one-to-one table joining. As done before, prospective similar addresses will undergo a longitude/latitude similarity test. Those that fail will be condensed by unique value rather than by the similar, matched address.

While evaluating the outcomes of **Step 1** applied to the prototype, it was identified that some addresses sharing the same `address_line_1` but differing in other metadata, such as city or five-digit zip code, were flagged for separation due to a failed geolocation test. It is assumed that the likelihood of a business filing under the exact same `address_line_1` in completely different locations is low enough to treat these as the same location. Therefore, such addresses will be marked for geolocation test override, and candidate matches will be determined by unique `address_line_1` and `city` combinations.

Appendix

`summarize_many_code_ranges_dt()` Output

For the implementation, refer to **SUBSECTION C1: Supplementary Location Metadata** and **SUBSECTION C2: Metadata Not Contributive to Project Goals** in “[Code/Process Data Update.R](#)”. The data processed in this step focused on a random subset of ABIs, and is only intended to illustrate metadata characteristics that require special attention.

Results over `vars_for_loc` were written to “./Data/Results/KEEP LOCAL/From Process Data Update/Area Code QC_Collapsed Data_05.18.2026.csv”. Results over `vars_to_sum` were written to “./Data/Results/KEEP LOCAL/From Process Data Update/Summarize Metadata Method_Collapsed Data_05.15.2026.csv”.

Table 8: Codebook for Evaluation Output Fields

Field	Description
<code>_ranges_OK</code>	Logical. TRUE when the <code>*_ranges</code> field (e.g., <code>census_tract_ranges</code>) assigns a value for every year listed in <code>available_year_ranges</code> , has no overlapping year conflicts (a year mapped to more than one distinct value), and all available years map to exactly one unique value overall. NA when <code>available_year_ranges</code> is missing or cannot be parsed.
<code>_ranges_decade_ok</code>	Logical or NA. Only evaluated when <code>*_ranges_ok</code> is FALSE. TRUE when, within each decade that has any mapped available years, all mapped years in that decade share the same value (i.e., ≤ 1 distinct value per decade). This check ignores gaps (missing years) within decades. NA when <code>*_ranges_ok</code> is TRUE or when <code>available_year_ranges</code> is missing/unparseable.
<code>_values_seen</code>	Character. Comma-separated list of distinct values observed for available years in the <code>*_ranges</code> mapping (restricted to years in <code>available_year_ranges</code>). Empty string if none are observed.
<code>_missing_years</code>	Character. Years present in <code>available_year_ranges</code> that are not covered by the <code>*_ranges</code> mapping. Empty string if none are missing. NA when <code>available_year_ranges</code> is missing/unparseable.

Continued on next page

Field	Description
<code>_conflict_years</code>	Character. Years that are assigned more than one distinct value due to overlapping segments in the <code>*_ranges</code> mapping, compressed as runs in the form "YYYY:YYYY" and separated by " , ". Empty string if no conflicts.

References

1. Ushey, K., Wickham, H. & Posit. [Introduction to renv](#).
2. Environmental Systems Research Institute (ESRI). Data axle business data dictionary. ArcGIS places documentation. <https://doc.arcgis.com/en/esri-demographics/latest/arcgis-places/data-axle.htm>.
3. U.S. Census Bureau. [2. Geographic boundaries, vintages, and frequency of updates](#). (2020).